

How to get to graduate school

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What you will need?

- Decide what do you want to study and **why**
- List of desired schools
 - *have a couple backup schools*
- Cover letter - why this particular school
- Essay
- 2-3 recommendation letters
- Transcript (often unofficial are good enough)
- list of papers, talks, posters, conferences
- which text books you used in high level classes
- money for the application fee (about \$50 each)
- GRE general and subject (depends on school)

What graduate admission looks for?

- Someone who is smart
 - *grades*
 - *competition prizes*
 - *but generally impossible to assess*
- Someone who **will get to finish line**
 - *knows why is a higher degree needed*
 - *knows the difference between undergrad and grad study*
 - *has a reasonable expectation*
 - *“go getter”*
 - *can survive in the face of unsolvable problem*
 - *you might get to a solution only after several years*
 - *or never*
- Someone who wants to be in this particular school
- brings/build community
 - *someone who is a good member of community*

- *i.e. not a lonely wolf*

Everything in your application should address the above points

How to choose subject area

- Do what ever *you* want but ...

Hard to get areas (subjective list)

- Astronomy (too popular)
- High energy (surprisingly popular but almost no jobs or grants)

Finding list of schools

- GRE requirements & admissions fees for US/Canadian Astronomy & Physics Programs tells you fees, GRE requirements
- Grad School Shopper quite clunky but still useful

Cover letter

- what do you want to achieve
- why this particular school
- what makes you good choice

Essay

Please, *PLEASE* do not start with: since I was 3 year old I was mesmerized by stars ...

- Give substance, expand cover letter
- Give specific examples of
 - *problems you faced*
 - *way to overcome*
 - *indication of grit*
- If you have problems with some grades
 - *explain why*
 - *what did you learn from that*
- What make you interesting person beside physics

Super important

- What skill set do you have
- Which particular subfield(s) you are interested
- Name couple profs - at least committee will know that you read something about school

Recommendation letters

- Ideally should add to the above topics
- Ideally from someone with whom you did research
 - *outside letters (REU/internship) are super good*
- Recent encounters, i.e. not from your kinder garden teacher
- Quite useless letter from a prof who saw you only in the class and never interacted with you
 - *we already know your grades*

How to ask for those?

- Just ask
- Talk with prof, describe your aspirations
- Give list of addresses/urls with **DEADLINES**
 - *if you can what subject of your interest*
 - *whom to address in the letter (if known)*

- once a letter is ready it costs nothing to send
 - *it is ok to ask for more*
 - *not ok to expect that it will be done right now*